Laboratory #4: Gravity Car

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Contents

1. Abstract
2. Model 0
3. Wheel Moment of Inertia Calculation
4. Model 1
5. Design & Analysis
6. Results & Discussion
7. References
8. Appendix



Bikini Bottom Bullet #42

1.0 Abstract

In this study, a gravity-powered vehicle, the Bikini Bottom Bullet #42, was designed and analyzed to predict its travel time along a multi-section track. Two theoretical models were developed to estimate performance. Model 0 treated the vehicle as an ideal point mass with no energy losses, while Model 1 incorporated additional physical effects including rotational inertia of the wheels and aerodynamic drag. Key parameters such as wheel moment of inertia were determined experimentally and incorporated into the model. The predicted travel times for Model 0 and Model 1 were 2.0814s and 2.2904 s, respectively, while the experimentally measured average time was 2.203 s. Model 1 reduced the relative error from 5.520% to 3.967%, demonstrating improved accuracy by accounting for non-ideal effects. These results highlight the importance of including realistic physical factors when modeling dynamic systems. This report begins with the derivation of Model 0, followed by the calculations used to determine the wheel moment of inertia. Model 1 is then developed, along with the corresponding free-body diagrams. The Design & Analysis section explains how these models informed the development of the Bikini Bottom Bullet. Finally, the results are presented and discussed.

2.0 Model 0

Model 0 serves as a baseline for analyzing the motion of the gravity-powered vehicle by considering an idealized system. In this model, the car is treated as a point mass, with all of its mass concentrated at a single location. To simplify the analysis, several assumptions are made, including a perfectly smooth track, no air resistance, and no energy losses due to friction. Under these conditions, all gravitational potential energy is converted directly into translational kinetic energy. The purpose of Model 0 is to establish a best-case scenario for the vehicle's performance and to provide a reference for evaluating more complex and realistic models. By analyzing the motion of the car along each section of the track using Newton's laws and energy methods, expressions for velocity and travel time can be derived. These results will later be compared to those of Model 1 and experimental data to assess the impact of non-ideal effects.

Assumptions:

- Perfect track (no bumps)
- No air resistance/drag
- No work from friction
- Car is a particle

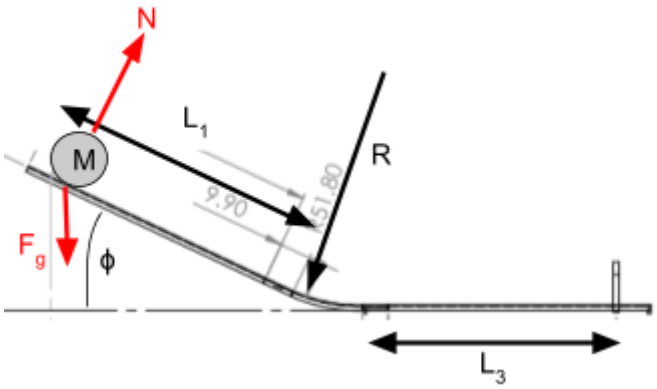


Figure 1. FBD for model 0

Consider L1 (angled section) as section 1 and L3 (flat section) as section 3. Section 2 will be the transfer section with the constant radius R.

For section 1:

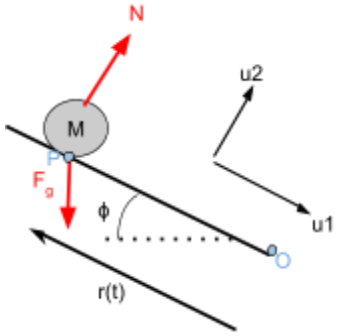


Figure 2. FBD for model 0, section 1

The kinematics for section 1 can be written as follows:

$$\vec{r(t)} = r(t) * \hat{u1}$$
$$\vec{v(t)} = \dot{r(t)} * \hat{u1}$$
$$\vec{a(t)} = \ddot{r(t)} * \hat{u1}$$

Next, we can sum the forces on the particle. The force of gravity is replaced with mg.

$$\Sigma\vec{F} = |\vec{N}| * \hat{u2} - |\vec{F_g}|cos\phi * \hat{u2} + |\vec{F_g}|sin\phi * \hat{u1}$$

Applying Newton's Second Law (f=ma) along $\hat{u2}$:

$$N = mgcos\phi$$

Along $\hat{u1}$:

$$\ddot{r(t)} = gsin\phi$$

Next, we assume that the particle starts at rest, and that the length that the particle travels during section 1 (L1), is the length from the starting gate to the transition section (section 2). Then, we can write the time that it takes for the particle to travel this section of the track using the kinematic equation:

$$\Delta x = v_0 t + 0.5at^2$$

Here, Δx is L1, v_0 is 0 m/s, and a is $gsin\phi$.

Rearranging for t and using the quadratic equation, the time spent traveling along section 1 comes out to be the expression:

$$t = \sqrt{\frac{2*L1}{gsin\phi}} \qquad (1)$$

The other equation needed is the final velocity of the particle during this section, since this is the initial velocity of the particle when it travels through the transfer section. The final velocity can be found using the following equation:

$$v_{f1}^2 = v_0^2 + 2a\Delta x$$

This equation simplifies to:

$$v_{f1} = \sqrt{2g * L1 * sin\phi} \qquad (2)$$

Next, we move to analyzing the particle's motion along the transfer section. For this analysis, we will use a N-T reference frame.

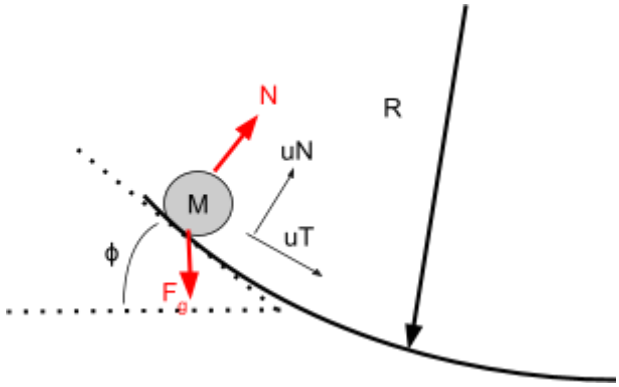


Figure 3. FBD for model 0, section 2 (transfer section)

For this section, we write the kinematics as follows:

$$\vec{v(t)} = \dot{s} * \hat{uT}$$

$$\vec{a(t)} = \ddot{s} * \hat{uT} + \frac{\dot{s}^2}{\rho} * \hat{uN}$$

The sum of forces on the particle is:

$$\Sigma\vec{F} = |\vec{N}| * \hat{uN} - |\vec{F_g}|cos\phi * \hat{uN} + |\vec{F_g}|sin\phi * \hat{uT}$$

Along $\hat{uT}$:

$$\ddot{s} = gsin\phi$$

We can write the following relations as well:

$$a = \frac{dv}{dt}$$

$$v = \frac{dv}{ds}$$

$$\frac{dv}{dt} = \frac{dv}{ds} * \frac{ds}{dt}$$

$$a \ = \ v * \frac{dv}{ds} = \ddot{s} = \ gsin\phi$$

The last relation we need is that $ds \ = \ Rd\phi$. Replacing this into $\frac{dv}{ds}$ gives us:

$$v * \frac{dv}{Rd\phi} = \ gsin\phi$$

$$v * dv \ = \ Rgsin\phi * d\phi$$

$$\int_{v_0}^{v_f} v * dv \ = \ gR \int_{\phi}^{0} sin\phi * d\phi$$

Solving for the final velocity gives us:

$$v_f = \sqrt{2Rg(1 \ - \ cos\phi) \ + v_{f1}^2}$$

Here, v_{f1} is the final velocity of the particle at the end of section 1. Replacing v_{f1} with the final velocity after section 1 gives us this equation:

$$v_{f2} = \sqrt{2Rg(1 \ - \ cos\phi) \ + \ (2gsin\phi \ * \ L1)} \qquad (3)$$

To calculate the time that the particle takes to travel this section, we need more relations:

$$v = \frac{ds}{dt}$$

$$dt \ = \ \frac{ds}{v}$$

$$ds \ = \ Rd\phi$$

$$\int dt = \int \frac{Rd\phi}{v}$$

$$t = \int_{0}^{\phi} \frac{Rd\theta}{\sqrt{2Rg(1-cos\theta)+(2gsin\phi*L1)}} \qquad (4)$$

Note that in equation 4, the $(2gsin\phi\ *\ L1)$ term is a constant. Since the final section is flat, and we are assuming no drag/air resistance or friction, there is no net force. This is shown in the figure below. No net force means that there is no acceleration, and so the particle travels at constant velocity, namely the final velocity of the particle after section 2 (equation 3).

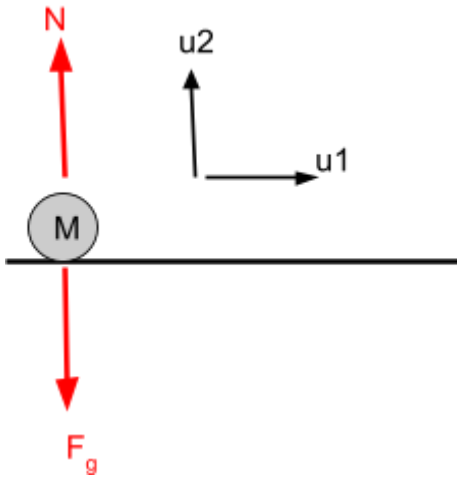


Figure 4. FBD for model 0, section 3

The sum of forces in this case is this:

$$\Sigma\vec{F} = |\vec{N}| * \hat{u2} - |\vec{F}_g| * \hat{u2}$$

$$N = F_g$$

There are zero forces along $\hat{u1}$ so there is no acceleration in that direction. Therefore, $d = v * t$ and so $t = \frac{d}{v} = \frac{L3}{v}$, where $v = \sqrt{2Rg(1 - cos\phi) + (2gsin\phi * L1)}$.

$$t = \frac{L3}{\sqrt{2Rg(1-cos\phi)+(2gsin\phi*L1)}} \qquad (5)$$

In summary, the total time for the particle to traverse the track is:

$$t = \sqrt{\frac{2*L1}{gsin\phi}} + \int_0^{\phi} \frac{Rd\theta}{\sqrt{2Rg(1-cos\theta)+(2gsin\phi*L1)}} + \frac{L3}{\sqrt{2Rg(1-cos\phi)+(2gsin\phi*L1)}} \qquad (6)$$

The table below summarizes model 0 with and without the parameters plugged in.

Table 1. Summary of model 0.

Track Section	Travel time (symbolically)	Travel time (seconds)

Section 1	$\sqrt{\frac{2*L1}{gsin\phi}}$	.979
Section 2	$\int_0^{\phi} \frac{Rd\theta}{\sqrt{2Rg(1-cos\theta)+(2gsin\phi*L1)}}$	0.170
Section 3	$\frac{L3}{\sqrt{2Rg(1-cos\phi)+(2gsin\phi*L1)}}$	0.936
Total	Eq. 6	**2.085**

3.0 Wheel Moment of Inertia Calculation

To determine the mass moment of inertia of the wheels experimentally, a pulley-based apparatus was constructed following the method described in the lab handout. A known hanging weight of W = 5.01 g = 0.04915 N was would around the pulley attachment at a string radius $r_{i,\,string}$, inducting a moment that caused the wheel to rotate about its axle. Two tests were performed: one with the wheel mounted on the pulley attachment, and one with the pulley attachment alone. The motion of the hanging weight was recorded using Vernier video analysis software, yielding a position-time data sampled at irregular intervals.

Applying Euler's rotational law about the wheel's center gives the equation of motion:

$$I_{total} \cdot \alpha = W \cdot r_{i,\,string} - F_f \cdot r_{i,\,axel}$$

where W = hanging weight (grams), $r_{i,\,string}$ = inner wheel radius (meters), $r_{i,\,axel}$ = inner axel radius (meters), $F_f = \mu \cdot N$ = Coulomb friction torque axel (μ is the Coulomb friction coefficient at the axle and N is the normal force at the axle), α is the angular acceleration, and $I_{total} = I_{wheel} + I_{pulley}$. Since both tests were performed using the same axle, weight, and string radius, the friction term is identical in each case. Writing the equation of motion for each test and subtracting eliminates friction entirely, yielding:

$$I_{wheel} \cdot \alpha_{wp} = I_{pulley} \cdot (\alpha_p - \alpha_{wp}).$$

To extract α from the video data, the vertical position of their hanging weight was fit to a quadratic of the form $y(t) = \frac{1}{2}at^2 + v_o t + y_o$ using least-squares regression. The constant linear acceleration a was then related to angular acceleration via the inextensible string constraint $\alpha = \frac{a}{r_{i,\,string}}$. Substituting this relation and simplifying $r_{i,\,string}$ cancels and the final expression for I_{wheel} is:

$$I_{wheel} = W \cdot r_{i,\,string}^{\,2}\left(\frac{1}{\alpha_p} - \frac{1}{\alpha_{wp}}\right),$$

where a_{wp} and a_p are the magnitudes of the linear accelerations from the wheel+pulley and pulley-only tests, respectively.

The quadratic regression applied to the vertical position data yielded a linear acceleration of a_{wp} = 3.94 m/s² for the wheel + pulley test and a_p = 7.17 m/s² for the pulley-only test. The quality of both fits was high, with $R^2 = 0.995$ for the wheel+pulley test and $R^2 = 0.994$ for the pulley-only test, confirming that the constant acceleration assumption inherent to the rigid body model is appropriate for this system. The notably lower acceleration in the wheel+pulley test relative to the pulley-only test is consistent with the additional rotational inertia introduced by the wheel resisting angular acceleration. To see each of the quadratic regression analyses, please refer to Figures 5 & 6 below.

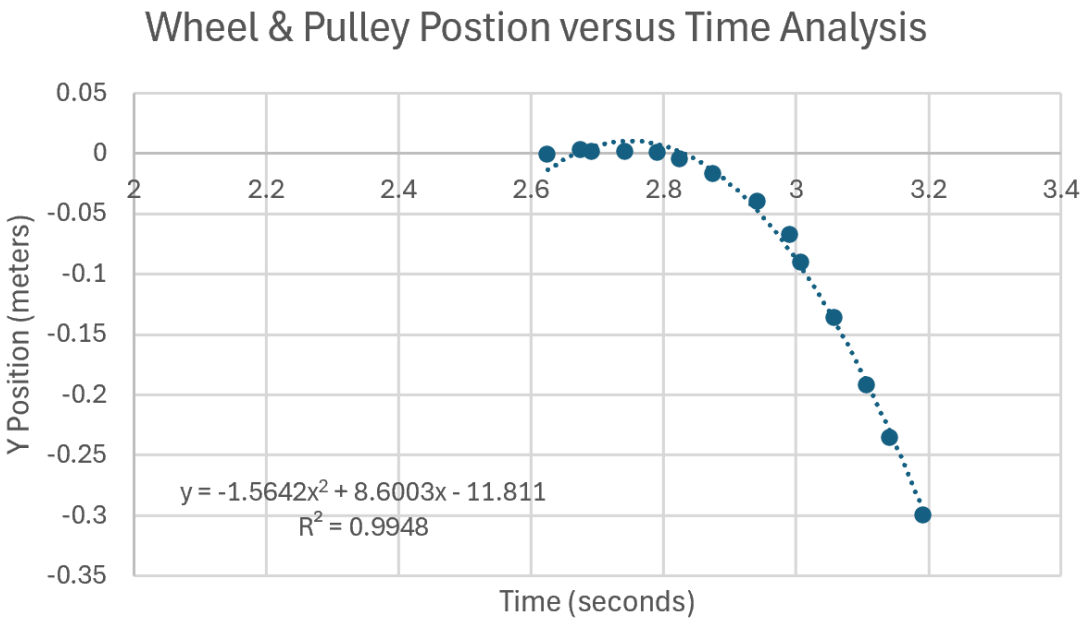


Figure 5. Wheel & Pulley Quadratic Regression Analysis

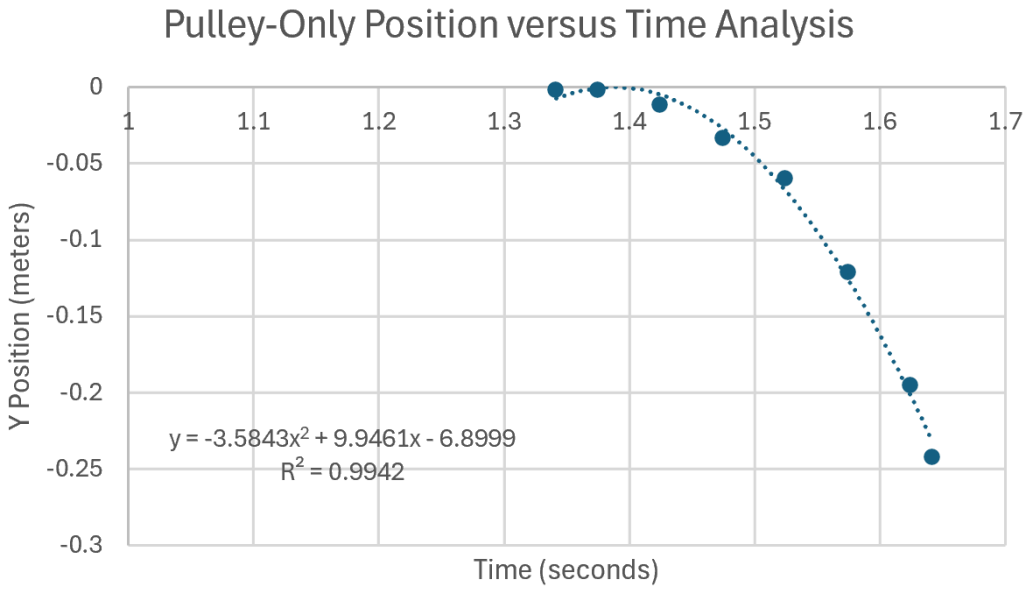


Figure 6. Pulley-Only Quadratic Regression Analysis

Substituting these values along with $W = 0.04915\ N$ into the derived expression gives:

$$I_{wheel} = W \cdot r_{i,string}{}^{2}\left(\frac{1}{\alpha_p} - \frac{1}{\alpha_{wp}}\right)$$

With a measured string radius of $r_{i,string}$ = 0.012505 m, this equation yields $I_{wheel} = 8.788 \times 10^{-7}\ kg \cdot m^2$. For comparison, the theoretical moment of inertia of a uniform disk is $I = \frac{1}{2}mr_o^2$, which with a wheel mass of 0.0014 kg and outer radius of 0.03175 m gives

$I_{theoretical} = 7.056 \times 10^{-7}\, kg \cdot m^2$. The agreement between the experimental and theoretical values is within reasonable experimental uncertainty, with a relative error of 24.5%. Any discrepancy may be attributed to the wheel not being a perfect uniform disk – for example, material removed at the hub or rim would shift the true moment of inertia away from the disk approximation – as well as minor vibration or slipping in the string during the experiment.

4.0 Model 1

Model 1 builds upon the idealized assumptions of Model 0 by incorporating additional physical effects to more accurately represent the behavior of the gravity-powered vehicle. In this model, the car is treated as a rigid body with four rotating wheels, rather than a point mass. As a result, both translational and rotational motion are considered, and a portion of the system's energy is distributed into the rotation of the wheels. Additionally, aerodynamic drag is included as a velocity-dependent force opposing the motion of the car.

Several assumptions are still maintained to keep the model tractable, including that the wheels roll without slipping and that the drag coefficient remains constant throughout the motion. Another assumption was that wheel-axle friction was negligible. By applying Newton's Second Law and accounting for these additional forces and energy contributions, a more realistic set of equations of motion is obtained. The results from Model 1 provide a more accurate prediction of the vehicle's performance and will be compared to both Model 0 and experimental data to evaluate the effectiveness of the model.

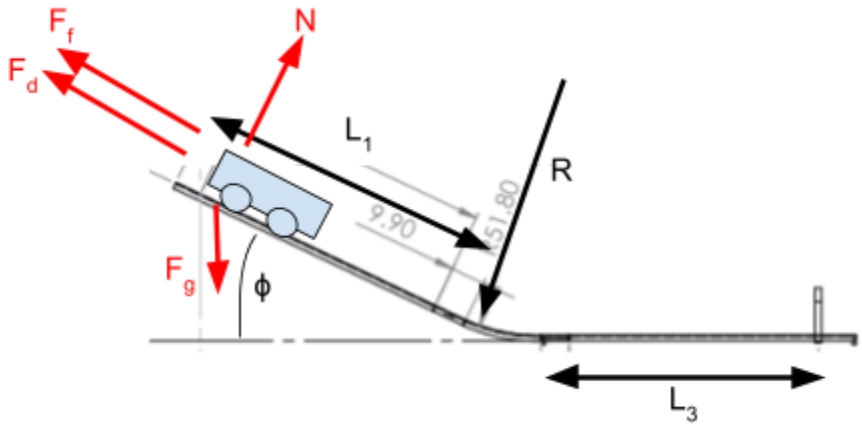


Figure 7. FBD for model 1

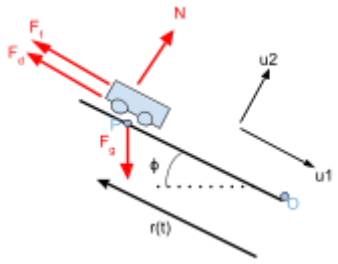


Figure 8. FBD of section 1 of track

The kinematics for section 1 can be written as follows:

$$\vec{r(t)} = r(t) * \hat{u1}$$

$$\vec{v(t)} = \dot{r(t)} * \hat{u1}$$
$$\vec{a(t)} = \ddot{r(t)} * \hat{u1}$$

Next, we can sum the forces on the particle in the u1 direction. The force of gravity is replaced with mg.

$$\Sigma\vec{F} = \ -\ |\vec{F_d}| * \hat{u1}\ -\ |\vec{F_f}| * \hat{u1}\ +\ |\vec{F_g}|sin\phi * \hat{u1}$$

Applying Newton's Second Law (f=ma) along $\hat{u1}$:

$$m_{car}a\ =\ \ -\ \vec{F_d}\ -\vec{F_f} + \vec{F_g}sin\phi$$

Next, to account for the force friction, we assume that the wheels are rolling without slipping.

$$T\ =\ F_f * r_{wheel} = I_{wheel} * \alpha$$

$$F_f = \frac{I_{wheel} * a}{r_{wheel}^2}$$

The air resistance experienced can be expressed as:

$$F_d = \ \frac{1}{2}C_d\rho A v^2$$

Putting these expressions together and rewriting Newton's Second law:

$$a * (m_{car} + 4\frac{I_{wheel}}{r_{wheel}^2}) = m_{car}gsin\phi\ -\frac{1}{2}C_d\rho A v^2$$

$$a\ = \frac{m_{car}gsin\phi - \frac{1}{2}C_d\rho A v^2}{m_{car} + 4\frac{I_{wheel}}{r_{wheel}^2}}$$

$$\text{Let } m_{eff} = m_{car} + 4\frac{I_{wheel}}{r_{wheel}^2} \text{ and } k = \frac{1}{2}C_d\rho A$$

This gives us two differential equations, which we can create an ODE system with.

$$\frac{dv}{dt} = \frac{m_{car}gsin\phi - kv^2}{m_{eff}}$$

$$\frac{dx}{dt} = v$$

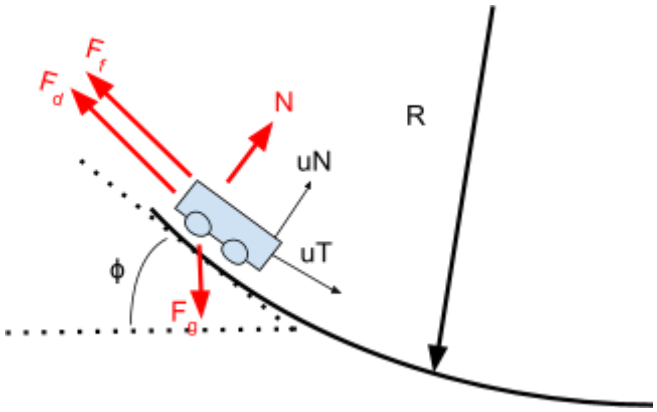


Figure 9. FBD of section 2 of model 1

For section two, we do the same but in a N-T reference frame.

$$\vec{v(t)} = \dot{s} * \hat{uT}$$

$$\vec{a(t)} = \ddot{s} * \hat{uT} + \frac{\dot{s}^2}{\rho} * \hat{uN}$$

The sum of forces on the particle in the tangential direction is:

$$\Sigma\vec{F} = |\vec{F_g}|sin\phi * \hat{uT} - |F_f| * \hat{uT} - |F_d| * \hat{uT}$$

Here, $F_f = \frac{I_{wheel} * \ddot{s}}{r^2_{wheel}}$ and $F_d = \frac{1}{2}C_d\rho A v^2$. We can replace $\frac{1}{2}C_d\rho A$ with k. Newton's second law shows that $m\ddot{s} = mgsin\phi - \frac{I_{wheel} * \ddot{s}}{r^2_{wheel}} - kv^2$. Let $m_{eff} = m_{car} + 4\frac{I_{wheel}}{r^2_{wheel}}$. Thus, our equation of motion system of ODEs comes out to be:

$$\ddot{s} = \frac{dv}{dt} = \frac{m_{car}gsin\phi - kv^2}{m_{eff}}$$

$$\frac{d\phi}{dt} = -\frac{v}{R}$$

Lastly, we do the same for section three.

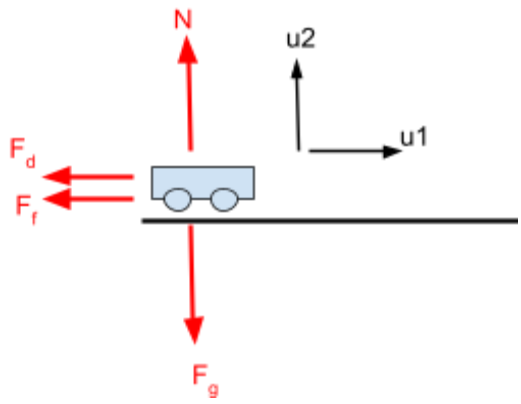


Figure 10. FBD of section 3 of model 1

The sum of forces in the u1 direction is:

$$\Sigma\vec{F} = -\ |F_d| - |F_f|$$

$$F_f = \frac{I_{wheel}{}^{*}a}{r_{wheel}^2}$$

$$F_d = \frac{1}{2}C_d\rho A v^2$$

Using Newton's second law, replacing $\frac{1}{2}C_d\rho A$ with k, and rearranging gives us that:

$$a = \frac{-kv^2}{m_{eff}}$$

Rewriting the EOM into a system of ODEs gives:

$$\frac{dv}{dt} = \frac{-kv^2}{m_{eff}}$$
$$\frac{dx}{dt} = v$$

This analysis results in systems of equations for each section of the track. A numerical solver (ode45) was used to solve for the final velocity during each section of the track and the time that the vehicle spent on that section of the track. This code is provided in appendix A.2.

The following table summarizes model 1:

Table 2. Summary of model 1.

Section of Track	ODE system	
Section 1 (downward ramp)	$\frac{dv}{dt} = \frac{m_{car} g sin\phi - kv^2}{m_{eff}}$	$\frac{dx}{dt} = v$
Section 2 (arc)	$\frac{dv}{dt} = \frac{m_{car} g sin\phi - kv^2}{m_{eff}}$	$\frac{d\phi}{dt} = -\frac{v}{R}$
Section 3 (flat)	$\frac{dv}{dt} = \frac{-kv^2}{m_{eff}}$	$\frac{dx}{dt} = v$

5.0 Design & Analysis

To begin our analysis, we need to calculate parameters of the track. The first will be the angle of the track at the starting position. A table of defined variables is presented below.

Table 3. Defined Variables to Calculate Parameters of Track

Variable	Representation [units]	Value
h_i	Starting height of the car [m]	1.105
h_f	Final height of the car after section 1 [m]	0.305
x	Width between starting height and final height of the car after section 1 [m]	1.77

To calculate the angle of the track, we can use the following formula:

$$\phi = tan^{-1}(\frac{h_i - h_f}{x})$$

Plugging in the values from the table, the angle of the track turns out to be 24.32°.

Next, we need to calculate the radius of the transfer section. To do that, 3 measurements were taken along the curve of the track with the point at which the transfer section meets the ground as the origin. These points are as follows (x, y):

P1 (0, 6.4)
P2 (101.5, 30.5)
P3 (50, 9.5)

Then, we calculate the distance between each point.

$$a = \overline{P1P2}$$
$$b = \overline{P2P3}$$
$$c = \overline{P3P1}$$

Next, we calculate the semiperimeter and area using these equations:

$$s = \frac{a+b+c}{2}$$

$$K = \sqrt{s(s-a)(s-b)(s-c)}$$

Finally, we use Herron's formula to calculate the radius of the curve:

$$R = \frac{a \cdot b \cdot c}{4K}$$

Plugging in all of the values, the radius of the curve is 1.63 m.

Design Features

The Bikini Bottom Bullet featured a curved body with four identical wheels. The body was designed with inspiration from modern day hypercars and features smooth curves throughout. During the design process, efforts were made to reduce the cross sectional area while covering the wheels and still maintaining smooth transitions between sections. The total length of the car was 177.80 mm and the total width of the car was 61.42 mm.

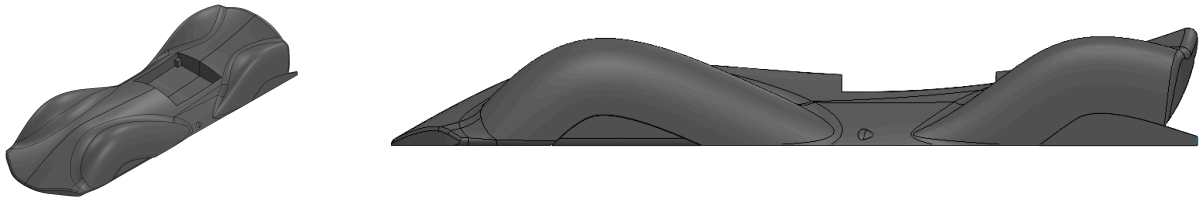


Figure 11. Isometric and side view of bodywork

A small square cutout was made in the bodywork to make space for additional weights that would be added later on. The wheels can be seen in Figure 12. The design for the wheels followed other popular designs to lower their moment of inertia by placing more mass near the axis of rotation. This design utilized 5 spokes to add strength due to the thin nature. The diameter of the center hole was decided on the availability of M3 bolts, which were used to mount the wheels to the bodywork. The outer diameter was chosen to be equivalent to standard pinewood derby wheels. The edge of the wheels have also been chamfered to decrease the size of the contact patch between the wheels and the ground to reduce rolling resistance. The trackwidth was chosen to maintain stability while preventing the track from interfering and accounting for manufacturing tolerances. Lastly, three degrees of negative camber was applied to each wheel to try and keep the vehicle stable while in motion.

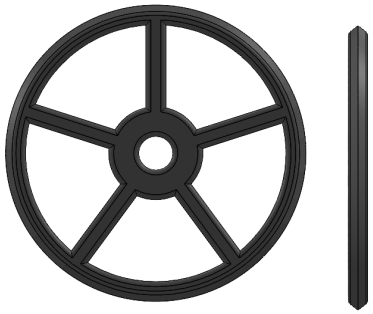


Figure 12. Front and side view of wheels

To manufacture, all parts of the car except for the weights and the bolts, were 3d printed. In assembly, holes were printed in the bodywork and a bolt was threaded through the holes with the wheels. This can be seen in Figure 13 (exploded view of the full assembly). Afterwards, a Philips head screwdriver was used to tune how tight the bolts pressed against the wheel. The bolts were tightened until the wheels were no longer wobbly, while still allowing the wheels to spin freely. Afterwards, WD40 was used.

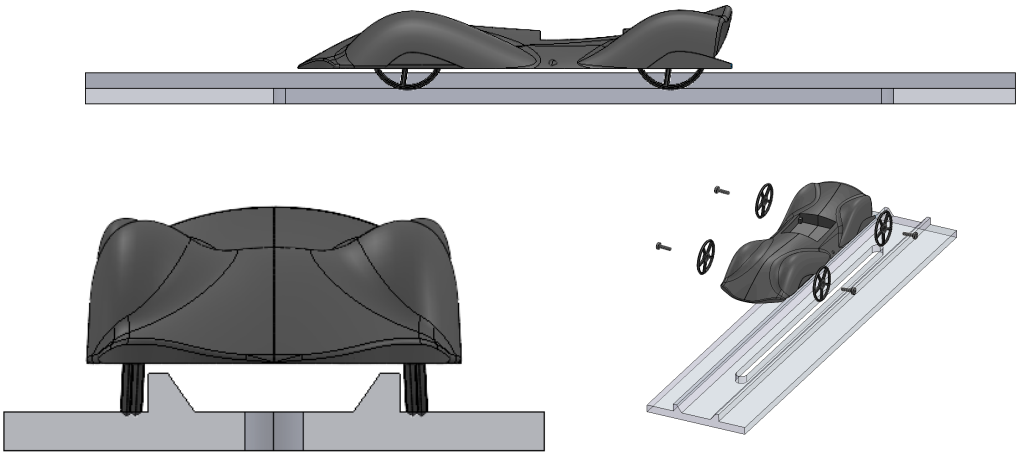


Figure 13. Side view, front view, and exploded view of the full assembly.

After manufacturing and initial testing, weight was added to the car. The car was also decorated using a colorful spongebob theme. The final car is shown in Figure 14.

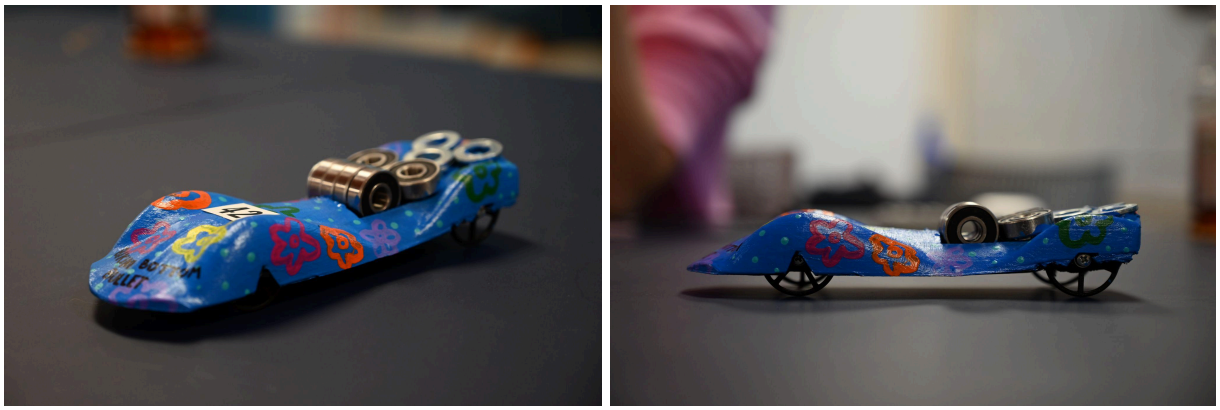


Figure 14. Isometric and side view of final car

6.0 Results & Discussion

This section presents the results obtained from the theoretical models and compares them to the experimental performance of the Bikini Bottom Bullet. Key parameters used in the models are summarized, followed by plots of velocity and position as functions of time for both Model 0 and Model 1. These results are used to evaluate how the inclusion of additional physical effects, such as rotational inertia and aerodynamic drag, influences the predicted motion of the vehicle. Finally, the predicted finish times are compared to the experimentally measured time, and potential sources of discrepancy between the models and real-world performance are discussed. All variables used in the models are summarized in Table 4.

Table 4. Table of parameters and their respective values

Variables	**Representation [units]**	**Value (if applicable)**
g	Gravitational acceleration [m/s^2]	9.81 m/s^2
L1	Length of angled section of track [m]	1.935 m
L3	Length of flat section of track [m]	4.024 m
R	Radius of transfer section of track [m]	1.63 m
m	Point mass representing the car for model 0 [kg]	0.137 kg
M	Mass of car [kg]	0.137 kg
m_{wheel}	Mass of wheel [kg]	0.0014 kg
r_{wheel}	Radius of wheel [m]	0.0149 m
ρ	Density of air [kg/m^3]	1.225 kg/m^3
Cd	Coefficient of drag of car [-]	0.5
A	Cross sectional area of car [m^2]	0.001375 m^2
ϕ	Angle from ground for the angled section of track [^{o}, rad]	24.32^o = 0.424 rad
$t_{model\ 0}$	Time [s]	2.0814 s
$t_{model\ 1}$	Time [s]	2.2904 s
t_{actual}	Time [s] – averaged across four trials	2.203 s
Δx	Distance traveled [m]	Dependent on section of track

First, to analyze the differences in Model 0 and Model 1, plots of velocity versus time and position along track were generated. To see the code used to generate these plots, please refer to the Appendix.

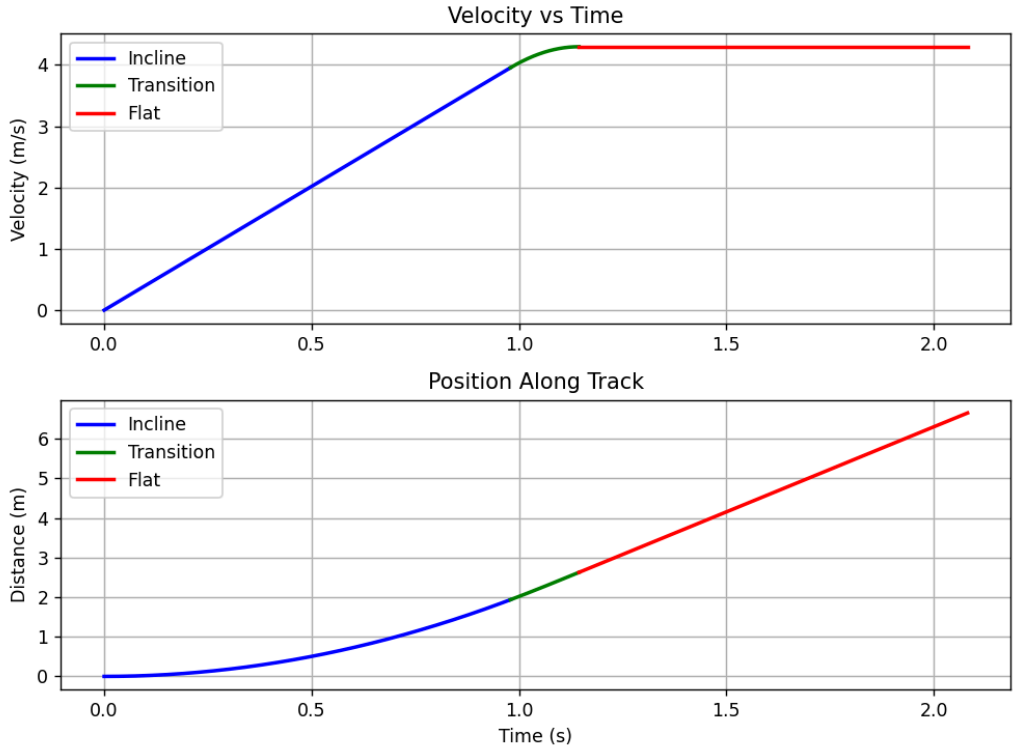


Figure 15. Model 0 Velocity and Position versus Time

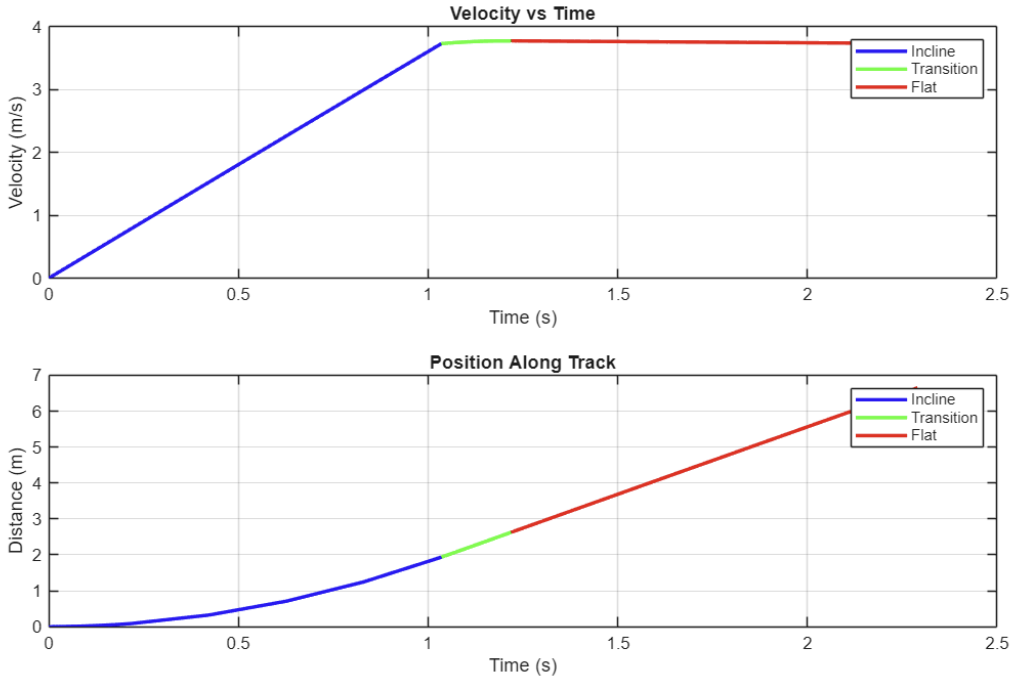


Figure 16. Model 1 Velocity and Position versus Time

Table 5. Specifics of Model 0 & Model 1

Model 0		**Model 1**	
Section 1 (Ramp)	End Velocity = 3.9521 m/s	Section 1 (Ramp)	End Velocity = 3.7318 m/s
	Time = 0.9792 s		Time =1.0352 s
Section 2 (Arc)	End Velocity = 4.2955 m/s	Section 2 (Arc)	End Velocity = 3.7763 m/s
	Time = 0.1654 s		Time = 0.2269 s
Section 3 (Flat)	End Velocity = 4.2955 m/s	Section 3 (Flat)	End Velocity = 3.7347 m/s
	Time = 0.9368 s		Time = 1.0715 s

The comparison between Model 0 and Model 1 demonstrates the effect of including more realistic physical behavior in the system. Model 0 assumes ideal motion with no energy losses, while Model 1 accounts for rotational inertia and aerodynamic drag. From the velocity vs. time plots, both models show similar trends, with velocity increasing along the incline, continuing to rise through the transition, and leveling off on the flat section. However, Model 1 consistently predicts lower velocities. At the end of the incline, Model 0 reaches 3.9521 m/s compared to 3.7318 m/s for Model 1. This difference becomes more pronounced in the transition section, where Model 0 reaches 4.2955 m/s while Model 1 only reaches 3.7763 m/s.

The time data also reflects this trend. Model 1 takes longer to traverse each section of the track, particularly on the flat portion, where drag continues to reduce velocity. While Model 0 maintains a constant velocity of 4.2955 m/s on the flat section, Model 1 speeds up slightly to 3.7763 m/s. These differences are also visible in the position vs. time plots, where Model 0 shows a steeper curve and reaches the end of the track faster. Overall, Model 0 overestimates performance due to neglecting energy losses, while Model 1 provides a more realistic prediction by accounting for drag and rotational effects, which reduce both velocity and overall acceleration.

The predicted times calculated for Model 0 and Model 1 were 2.0814 and 2.2904 seconds, respectively. The sum of the four trials for the Bikini Bottom Bullet was 8.8250, resulting in an average time of 2.203 seconds per trial. The table below presents the relative percent errors between the predicted times and actual time for our car.

Table 6. Percent Errors

	Model 0	Model 1
Bikini Bottom Bullet	5.520%	3.967%

The percent error results show a clear improvement in predictive accuracy from Model 0 (5.520%) to Model 1 (3.967%), indicating that increasing model complexity led to predictions that more closely matched the experimental performance of the Bikini Bottom Bullet. Model 0 underestimates the actual time because it assumes an idealized system with no energy losses and treats the car as a point mass. This neglects important real-world effects such as rotational inertia of the wheels and friction losses, which act to slow the car down and increase travel time. Model 1 reduces this discrepancy by incorporating rotational inertia and treating the car as a rigid body with rolling wheels. This added realism accounts for energy being distributed into both translational and rotational motion, which better reflects the physical behavior of the system and results in a lower percent error. However, Model 1 slightly overestimates the actual time, which suggests that although it captures the system more realistically, some parameters or assumptions may still be different from the true behavior of the car. The assumed drag coefficient and the cross-sectional area might lead to overestimated aerodynamic losses. One of the front wheels may not have been in contact with the track, and this could be where the model overestimated the rotational inertia and rolling behavior of the wheels.

These differences are expected because both models still rely on simplifying assumptions. For example, neither model likely fully captures friction in the axles, rolling resistance, slight misalignments in the track, or uncertainty in the estimated drag coefficient and cross-sectional area. Small experimental uncertainties – such as timing precision, variations between trials, or minor inconsistencies in release conditions – can also contribute to the observed deviation. Additional discrepancies may arise from axle friction, track imperfections, and uncertainty in parameter estimation. As noted in the lab manual, there is a tradeoff between model complexity and practical accuracy, and some level of disagreement is inevitable when simplifying a real physical system. The trend in percent error demonstrates that incorporating more realistic physical effects improves predictive capability but also highlights that even refined models will have limitations due to unmodeled losses and experimental uncertainty. Overall, Model 0 overestimates performance due to neglecting energy losses, while Model 1 provides a more realistic prediction by accounting for drag and rotational effects, which reduce both velocity and overall acceleration.

7.0 References

Douglas, C., Dowell, E., & McGuire, P. (2026). EGR 244L: Laboratory #4 – Gravity car. Duke University, Pratt School of Engineering.

8.0 Appendix

Appendix A.1. Solving Model 1's ramp section ODE analytically.

$$\frac{dv}{dt} = \frac{m_{car} g sin\phi - kv^2}{m_{eff}}$$

Rewriting this and replacing the constants gives:

$$\frac{dv}{dt} = A - Bv^2$$

$$A = \frac{m_{car} g sin\phi}{m_{eff}}$$

$$B = \frac{k}{m_{eff}}$$

Solving the differential equation using separation of variables gives:

$$v(t) = \sqrt{\frac{A}{B}} tanh(\sqrt{AB}t)$$

Now, the time for the car to travel down section one of the track has to be calculated. This can be done by finding the velocity of the car at the end of this section of the track.

$$a = v \cdot \frac{dv}{ds} = A - Bv^2$$

Rearranging and integrating to solve for *s*:

$$s = \frac{1}{2B} ln(\frac{A}{A - Bv^2})$$

Solving for *v* gives:

$$v = \sqrt{\frac{A}{B}} \cdot \sqrt{1 - e^{-2Bs}}$$

Plugging in $L1 = s$ gives us the final velocity of the car after section 1.

$$v_{f1} = \sqrt{\frac{A}{B}} \cdot \sqrt{1 - e^{-2B \cdot L1}}$$

To calculate the time required for the car to travel through section 1, $v(t) = \sqrt{\frac{A}{B}}tanh(\sqrt{AB}t)$ can be rearranged for t.

$$t = \frac{1}{\sqrt{AB}} \cdot tanh^{-1}(\frac{v_{f1}}{\sqrt{\frac{A}{B}}})$$

Appendix A.2. Code used to analyze model 1

```
clc;
clf;

%parameters

g = 9.81; %m/s^2
L1 = 1.935; %m
L3 = 4.024; %m
R = 1.63; %m

phi = 0.424;

m = 0.137; %kg
I = 8.788e-7; %kg*m^2
r = 0.0149; %m

rho = 1.225; %kg/m^3
Cd = 0.5; %guess
A = 0.001375; %m^2

k = 0.5 * rho * Cd * A;

m_eff = m + 4*I/r^2;

%section 1

ode_section1 = @(t,y) [(m*g*sin(phi) - k*y(1)^2)/m_eff; y(1)];
```

```
function [value, isterminal, direction] = event1(t,y,L1)
   value = y(2) - L1;
   isterminal = 1;
   direction = 1;
end
```

```
opts1 = odeset('Events', @(t,y) event1(t,y,L1));
```

```
[t1, y1] = ode45(ode_section1, [0 10], [0; 0], opts1);
```

```
v1_end = y1(end,1);
t1_end = t1(end);
```

```
fprintf('Section 1 (ramp)\n');
fprintf('end velocity: %.4f m/s\n', v1_end);
fprintf('time: %.4f s\n\n', t1_end);
```

```
%section 2
```

```
theta0 = phi;
```

```
ode_section2 = @(t,y) [m*g*sin(y(2)) - k*y(1)^2/m_eff; -y(1)/R];
```

```
function [value, isterminal, direction] = event2(t,y)
   value = y(2);     %stops when angle becomes 0
   isterminal = 1;
```

```
   direction = -1;
end

opts2 = odeset('Events', @event2);

[t2, y2] = ode45(ode_section2, [0 10], [v1_end; theta0], opts2);

v2_end = y2(end,1);
t2_end = t2(end);

fprintf('Section 2 (arc)\n');
fprintf('end velocity: %.4f m/s\n', v2_end);
fprintf('time: %.4f s\n\n', t2_end);

%section 3

ode_section3 = @(t,y) [(-k*y(1)^2)/m_eff;y(1)];

function [value, isterminal, direction] = event3(t,y,L3)
   value = y(2) - L3;
   isterminal = 1;
   direction = 1;
end

opts3 = odeset('Events', @(t,y) event3(t,y,L3));

[t3, y3] = ode45(ode_section3, [0 20], [v2_end; 0], opts3);
```

```
t3_end = t3(end);

fprintf('Section 3 (Flat)\n');
fprintf('final velocity: %.4f m/s\n', y3(end,1));
fprintf('fime: %.4f s\n\n', t3_end);

%summing all sections
t_total = t1_end + t2_end + t3_end;

fprintf('TOTAL TRACK TIME\n');
fprintf('Total time: %.4f s\n', t_total);



figure;

%velocity plot
subplot(2,1,1)
plot(t1, y1(:,1), 'b', 'LineWidth', 2); hold on;
plot(t1_end + t2, y2(:,1), 'g', 'LineWidth', 2);
plot(t1_end + t2_end + t3, y3(:,1), 'r', 'LineWidth', 2);
xlabel('Time (s)');
ylabel('Velocity (m/s)');
legend('Incline','Transition','Flat');
title('Velocity vs Time');
grid on;

%position plot
```

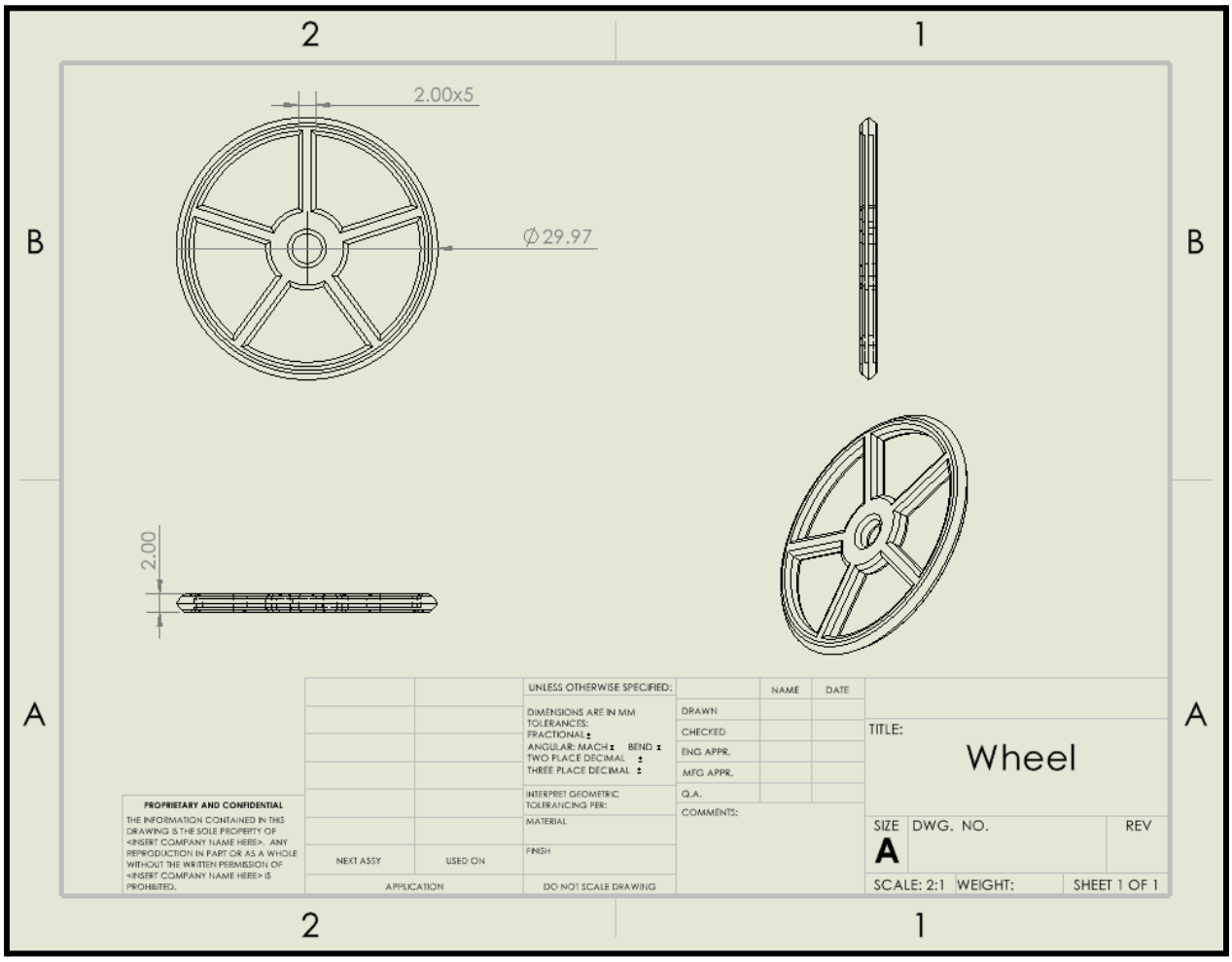
```
subplot(2,1,2)
plot(t1, y1(:,2), 'b', 'LineWidth', 2); hold on;

theta = y2(:,2);
s_arc = R*(theta0 - theta);
plot(t1_end + t2, L1 + s_arc, 'g', 'LineWidth', 2);

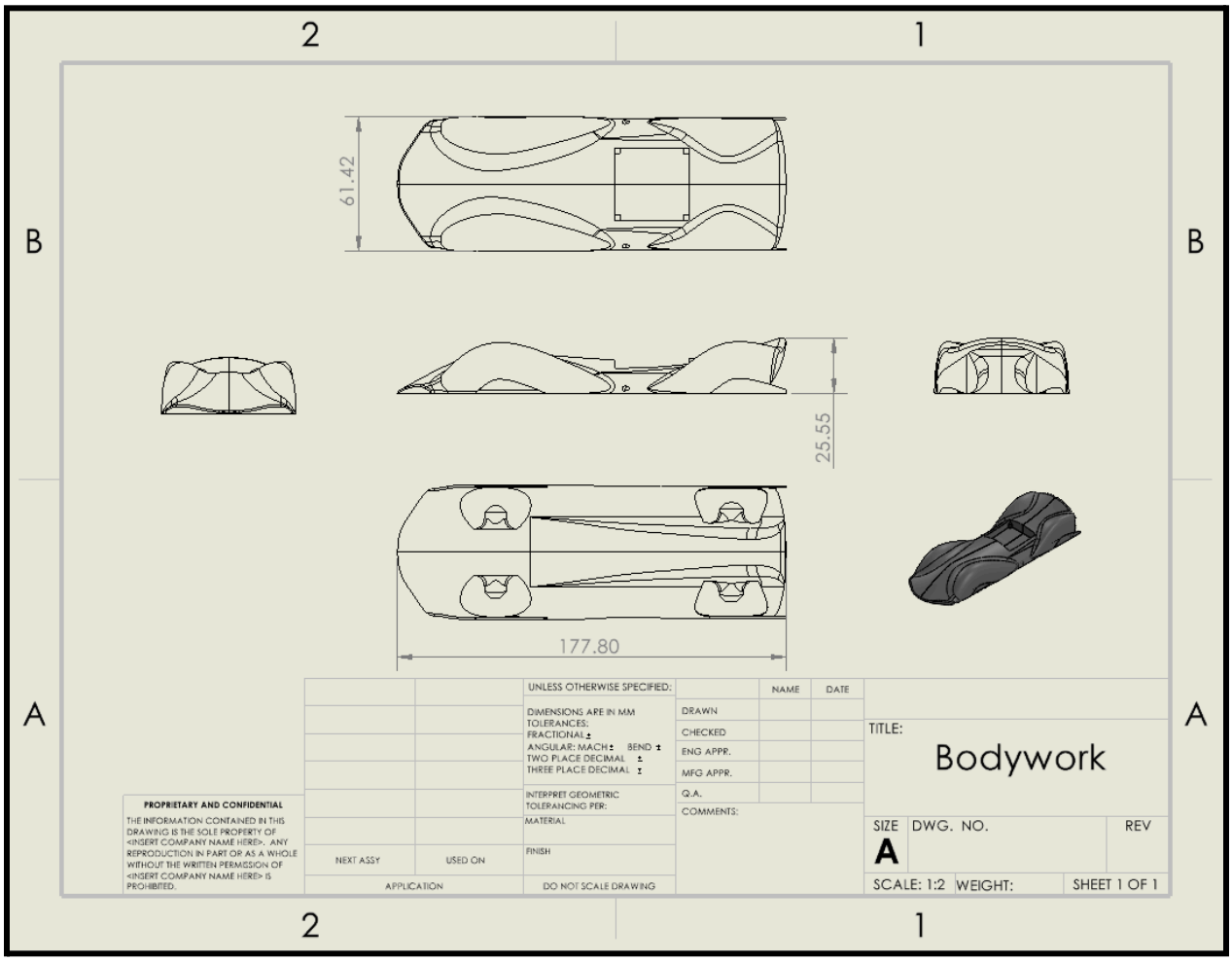
plot(t1_end + t2_end + t3, L1 + s_arc(end) + y3(:,2), 'r', 'LineWidth',
2);

xlabel('Time (s)');
ylabel('Distance (m)');
legend('Incline','Transition','Flat');
title('Position Along Track');
grid on;
```

Appendix A.3. Drawing of Wheel



Appendix A.4. Drawing of Bodywork



Appendix A.5. Code used to analyze model 0

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.integrate import solve_ivp

# PARAMETERS
g = 9.81
L1 = 1.935
L3 = 4.024
R  = 1.63
phi = 0.424

# SECTION 1 (Incline)
def section1(t, y):
    v, x = y
    return [g*np.sin(phi), v]

def event1(t, y):
    return y[1] - L1
event1.terminal = True
event1.direction = 1

sol1 = solve_ivp(section1, [0,10], [0,0], events=event1, max_step=0.001)

t1 = sol1.t
v1 = sol1.y[0]
x1 = sol1.y[1]

v1_end = v1[-1]
t1_end = t1[-1]

print("Section 1 (ramp)")
print(f"end velocity: {v1_end:.4f} m/s")
print(f"time: {t1_end:.4f} s\n")

# SECTION 2 (Arc)
def section2(t, y):
    v, theta = y
    return [g*np.sin(theta), -v/R]

def event2(t, y):
    return y[1]
event2.terminal = True
event2.direction = -1

sol2 = solve_ivp(section2, [0,10], [v1_end, phi], events=event2, max_step=0.001)

t2 = sol2.t
```

```
v2 = sol2.y[0]
theta = sol2.y[1]

t2_shift = t1_end + t2

s_arc = R * (phi - theta)
x2 = L1 + s_arc

v2_end = v2[-1]
t2_end = t2[-1]

print("Section 2 (arc)")
print(f"end velocity: {v2_end:.4f} m/s")
print(f"time: {t2_end:.4f} s\n")

# SECTION 3 (Flat)
def section3(t, y):
    v, x = y
    return [0, v]

def event3(t, y):
    return y[1] - L3
event3.terminal = True
event3.direction = 1

sol3 = solve_ivp(section3, [0,20], [v2_end, 0], events=event3, max_step=0.001)

t3 = sol3.t
v3 = sol3.y[0]
x3 = sol3.y[1]

t3_shift = t1_end + t2_end + t3
x3_total = L1 + s_arc[-1] + x3

t3_end = t3[-1]

print("Section 3 (Flat)")
print(f"final velocity: {v3[-1]:.4f} m/s")
print(f"time: {t3_end:.4f} s\n")

# TOTAL TIME
t_total = t1_end + t2_end + t3_end

print("TOTAL TRACK TIME")
print(f"Total time: {t_total:.4f} s")

# PLOTS
plt.figure(figsize=(8,6))
```

```
# Velocity plot
plt.subplot(2,1,1)
plt.plot(t1, v1, 'b', linewidth=2)
plt.plot(t2_shift, v2, 'g', linewidth=2)
plt.plot(t3_shift, v3, 'r', linewidth=2)
plt.ylabel("Velocity (m/s)")
plt.title("Velocity vs Time")
plt.legend(['Incline','Transition','Flat'])
plt.grid()

# Position plot
plt.subplot(2,1,2)
plt.plot(t1, x1, 'b', linewidth=2)
plt.plot(t2_shift, x2, 'g', linewidth=2)
plt.plot(t3_shift, x3_total, 'r', linewidth=2)
plt.xlabel("Time (s)")
plt.ylabel("Distance (m)")
plt.title("Position Along Track")
plt.legend(['Incline','Transition','Flat'])
plt.grid()

plt.tight_layout()
plt.show()
```